

Partial Algebraic Shifting

Antony Della Vecchia

Joint work with F. Lenzen and M. Joswig

TU Berlin

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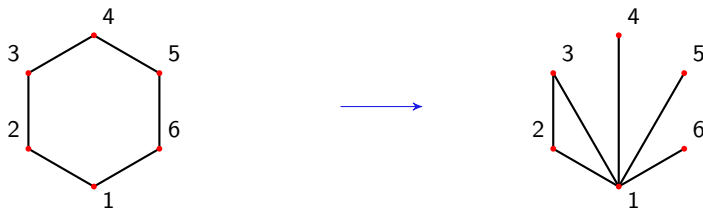
Algebraic Shifting

Definition

$C \in \binom{[n]}{k}$ is shifted if for any $\sigma \in C$, $\sigma|_i^j \in C$ for $i \in \sigma$ and $j < i$.

Theorem (Kalai, Björner-Kalai)

Given K a simplicial complex and a field $\mathbb{F}$, there exists a shifted simplicial complex $\Delta^{\mathbb{F}}(K)$ with the same f -vector and betti numbers as K .



Exterior Algebraic Shifting

Definition

Let $\mathbb{E} \supset \mathbb{F}$ be a field extension and let $g \in \text{GL}(\mathbb{E}, n)$.

- ▶ The k th compound matrix $g^{\wedge k}$ is the $\binom{n}{k} \times \binom{n}{k}$ -matrix of k -minors of g .
- ▶ $g^{\wedge K_k}$ is the $|K_k| \times \binom{n}{k}$ -submatrix of $g^{\wedge k}$ with rows indexed by K_k .

Definition

The partial (exterior) shift of K by g is

$$\Delta_g^{\mathbb{F}}(K) := \bigcup_{k \leq \dim(K)} \left\{ \sigma \in \binom{[n]}{k} : g_{*\sigma}^{\wedge K_k} \notin \text{span}_{\mathbb{E}}((g_{*\rho}^{\wedge K_k})_{\rho <_{\text{lex}} \sigma}) \right\}.$$

For $g = (x_{ij})_{1 \leq i, j \leq 3}$ and $K = \{\{1, 2\}, \{2, 3\}\}$.

$$g^{\wedge 2} = \begin{pmatrix} x_{11}x_{22} - x_{12}x_{21} & x_{11}x_{23} - x_{13}x_{21} & x_{12}x_{23} - x_{22}x_{13} \\ x_{21}x_{32} - x_{22}x_{31} & x_{21}x_{33} - x_{23}x_{31} & x_{22}x_{33} - x_{23}x_{32} \end{pmatrix},$$

$\Delta(K) = \{\{1, 2\}, \{1, 3\}\}$ mention monte carlo



Algebraic Exterior Shifting

$$e_\sigma = e_{\sigma_1} \wedge \dots \wedge e_{\sigma_k} \text{ for } \sigma = \{\sigma_1 < \dots < \sigma_k\}.$$

$$I := (e_\sigma : \sigma \notin K) \subset \bigwedge \mathbb{E}^n, I_{\Delta_g(K)} = \text{in}(g \cdot I)$$

plucker?



Partial Algebraic Shifting

Consider now $\{\tau(w) \in \mathrm{GL}(\mathbb{E}, n) : w \in S_n\}$.

Let $c_n := (1 \dots n) \in S_n$.

Theorem (DV-Joswig-Lenzen)

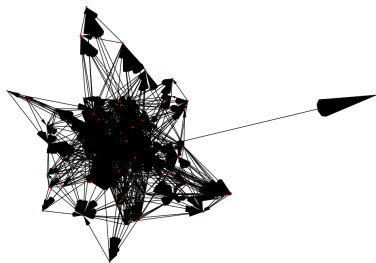
For any simplicial complex K and $w \in S_n$ such that $c_n \leq w$ in the weak Bruhat order we have the following,

1. $\Delta_{\tau(w)}(K)$ is a homotopy equivalent to a wedge of spheres.
2. $\Delta_{\tau(w)}(K)$ has the same Betti numbers as K .



Real Projective Plane Example

$$K = \mathbb{RP}_6^2, \mathbb{F} = \mathbb{Q}$$



- ▶ show example of how we can compute full shift of projective plane with standard n cycle
- ▶ computations inspired us to find result
- ▶ example shows that our result is tight



LU decomposition

Lemma

$\Delta_{gb}(K) = \Delta_g(K)$ for any upper triangular b and any simplicial complex K .

LU decomposition For $g \in \text{GL}(n, \mathbb{E})$ with non zero leading principal minors, there exists l lower triangular, u upper triangular such that

$$g = lu$$

For generic $g \in \text{GL}(n, \mathbb{E})$, $\Delta_l(K) = \Delta_g(K)$.

I.e. we can use l with $\frac{1}{2}n(n-1)$ transcendentals.



Inversions and Rothe matrix

Definition

For $w \in S_n$, let

$$\text{Inv } w := \{(i, j) : i < j, i \cdot w > j \cdot w\}$$

$$\ell(w) := |\text{Inv } w|.$$

$$u(w) := \begin{pmatrix} 1 & u_{12} & \cdots & u_{1n} \\ & \ddots & \ddots & \vdots \\ & & 1 & u_{n-1,n} \\ & & & 1 \end{pmatrix} \quad \text{with} \quad u_{ij} = \begin{cases} x_{ij} & \text{if } (i, j) \in \text{Inv } w, \\ 0 & \text{otherwise,} \end{cases}$$

and let $\tau(w) := u(w)w$.

Example

Take $w = (2 \ 3)$, $\text{Inv } w = \{(1, 2), (1, 3)\}$.

$$\begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & y & x \\ & z & 1 \\ & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & y & x \\ & 1 & 0 \\ & & 1 \end{pmatrix}.$$



Bruhat decomposition

Let $B \subset \mathrm{GL}(n, \mathbb{E})$ be the subgroup of upper triangular matrices,

$$\mathrm{GL}(n, \mathbb{E}) = \coprod_{w \in \mathrm{Sym}(n)} BwB .$$

Describe w_0

$$\Delta_{w_0 g}(K) = \Delta_g(K)$$

$w_0 g = lu = w_0 u' w_0 u$ (bring the two decompositions together)



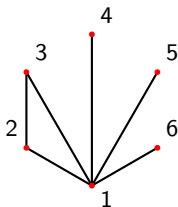
Near cones

Definition

A simplicial complex K is a near cone if for every $\sigma \in K$ with $1 \notin \sigma$ we have $\sigma|_i^1 \in K$ for each $i \in \sigma$.

Lemma (Björner-Kalai 88)

Let K be a near cone. Then K is homotopy equivalent to a wedge of spheres and $\beta_k(K) = |\{\sigma \in K^k : \sigma \cup \{1\} \notin K\}|$.



Shifting to near cones

Lemma (DV-Joswig-Lenzen)

Let K be a simplicial complex on n vertices. Further, let $g \in \mathrm{GL}(n, \mathbb{E})$ be a matrix whose entries in the first column are algebraically independent over $\mathbb{F}$ and the other entries of g . Then $\Delta_g(K)$ is a near cone.

Example

The n -cycle $c_n := (1 \ 2 \ \dots \ n)$ gives rise to the matrix

$$\tau(c_n) = \begin{pmatrix} x_{1,n} & 1 & & \\ \vdots & & \ddots & \\ x_{n-1,n} & & & 1 \\ 1 & 0 & \dots & 0 \end{pmatrix}.$$

$$\begin{aligned} \Delta_{\tau(c_6)}^{\mathbb{Q}}(\mathbb{RP}_6^2) &= \Delta\{123, 124, 125, 126, 134, 135, 136, 145, 146, 156\} \\ &= \Delta^{\mathbb{Q}}(\mathbb{RP}_6^2) \end{aligned}$$



Shifting to near cones

Lemma (Björner-Kalai)

Let $g \in \text{GL}(n, \mathbb{E})$ be generic and suppose for $\tau \in \binom{[n]}{k}$ there exists $\gamma \in \mathbb{E}$ such that,

$$g_{\tau}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma_{\sigma} g_{\sigma}^{\wedge K_k}.$$

If $w \in S_n$, then there exists $\gamma' \in \mathbb{E}$ such that

$$g_{\tau \cdot w}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma'_{\sigma \cdot w} g_{\sigma \cdot w}^{\wedge K_k}$$

spell out connection with generic first column shifting to near cone



Weak Bruhat order

- ▶ definition and equivalent formulation
- ▶ an example with rothe matrix that shows shifting to near cones



Proof of results

Theorem (DV-Joswig-Lenzen)

For K a simplicial complex, $v, w \in S_n$ with $\ell(vw) = \ell(v) + \ell(w)$,

$$\Delta_{\tau(vw)}(K) = \Delta_{\tau(v) \tau(w)^v}(K)$$



Mardi slide?

- ▶ talk about how file format was useful



QR codes to both papers
Thank You!

